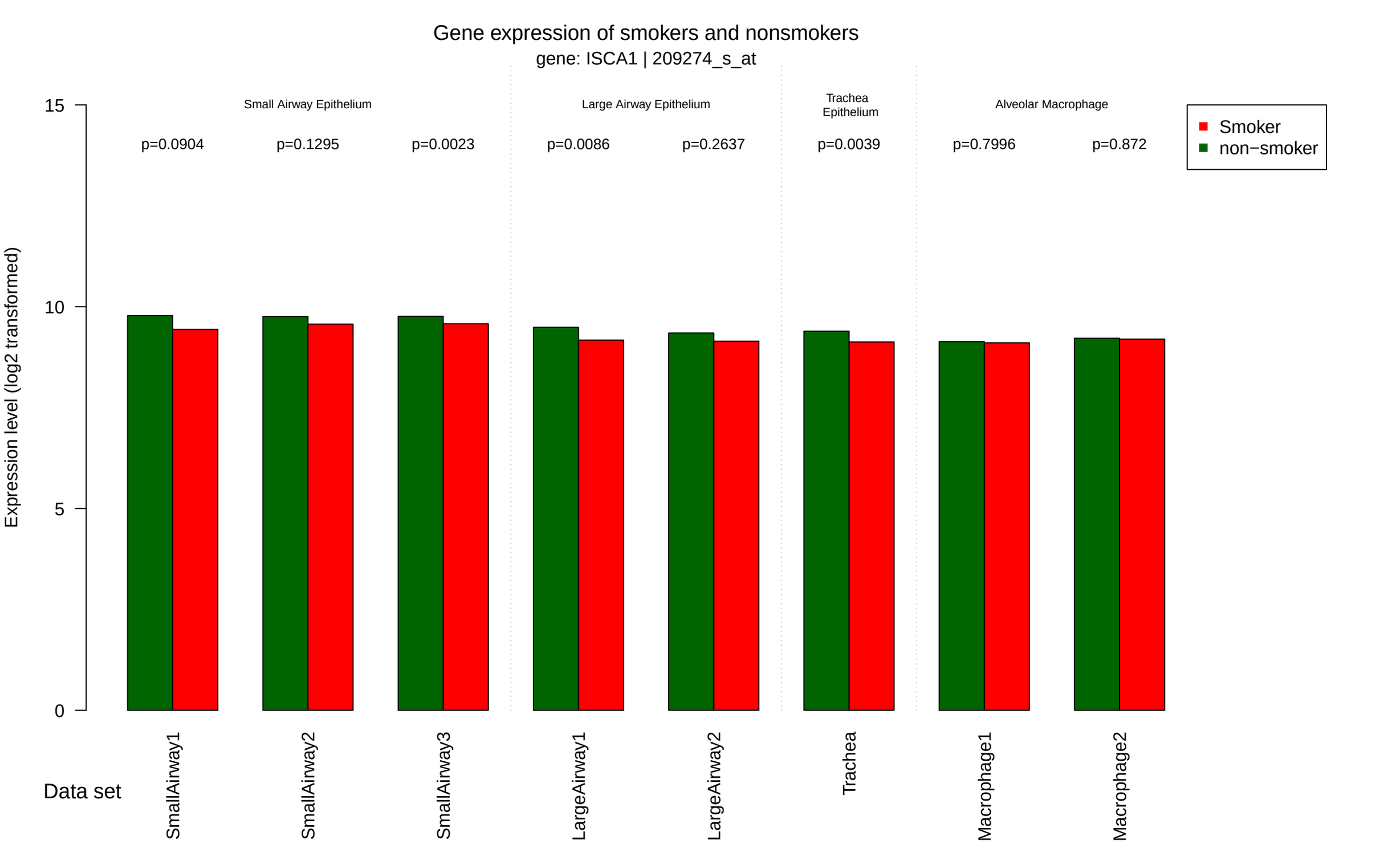
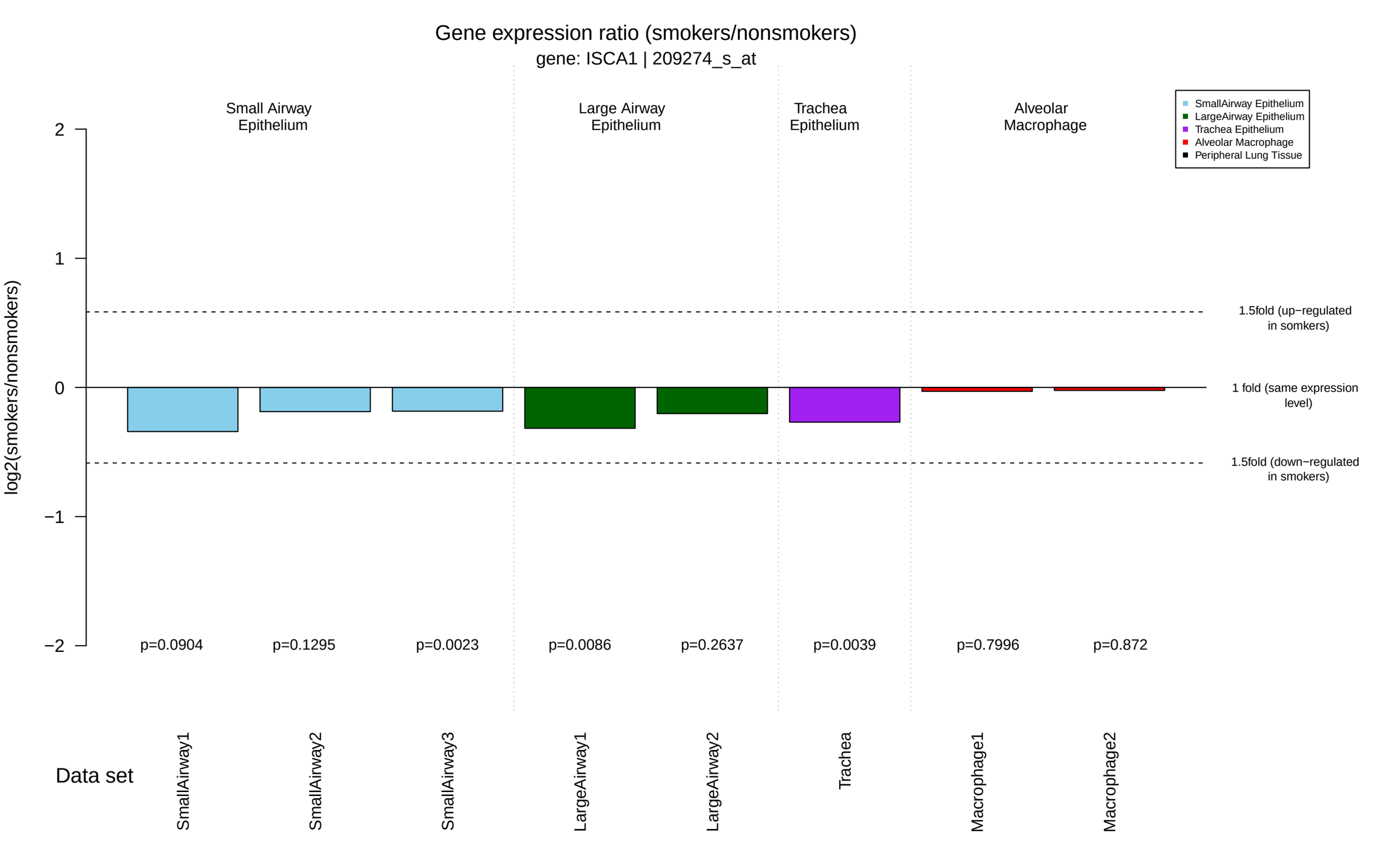
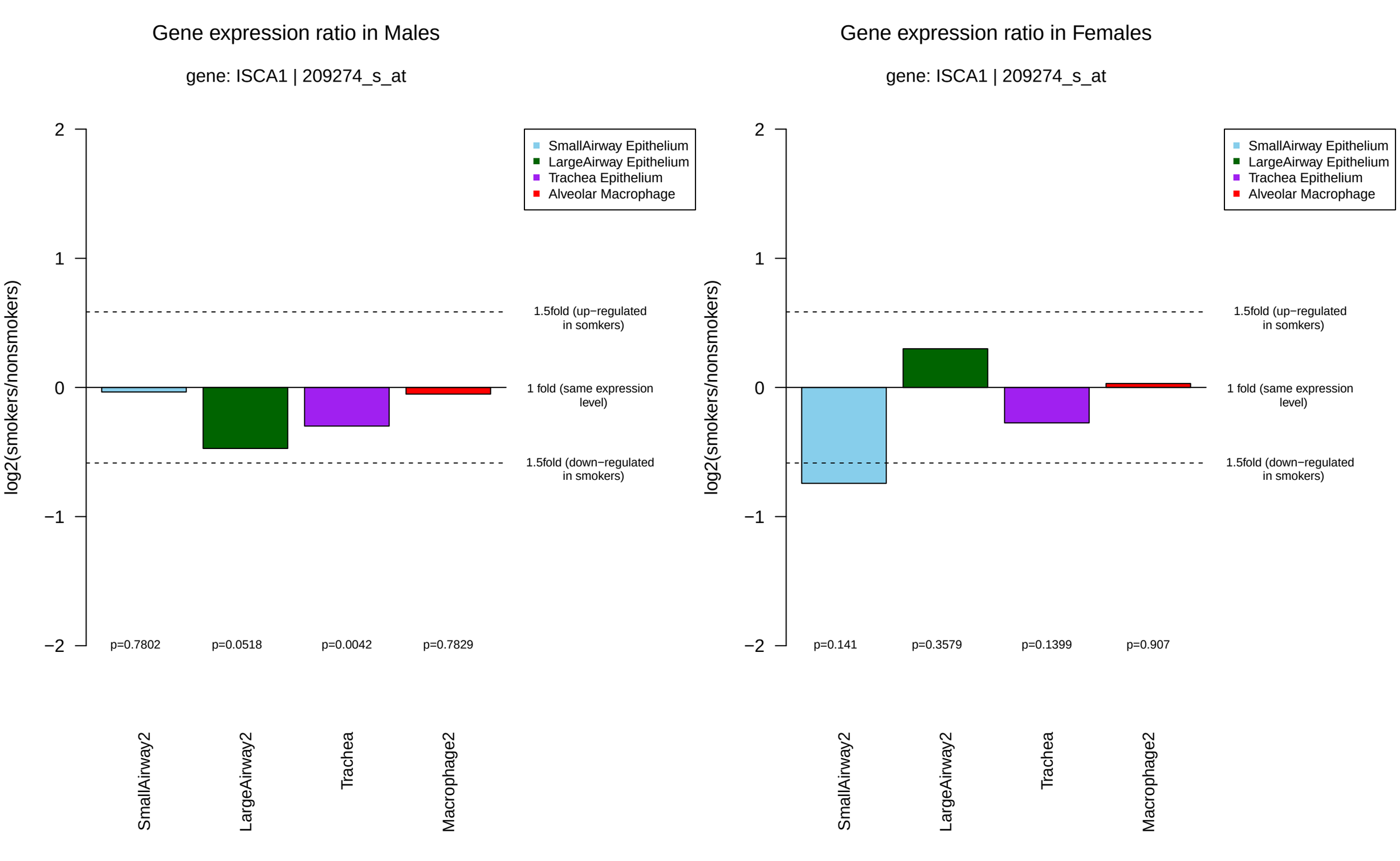
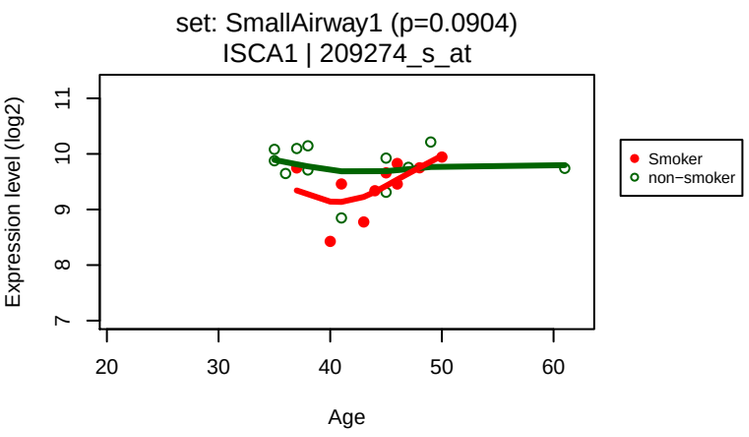
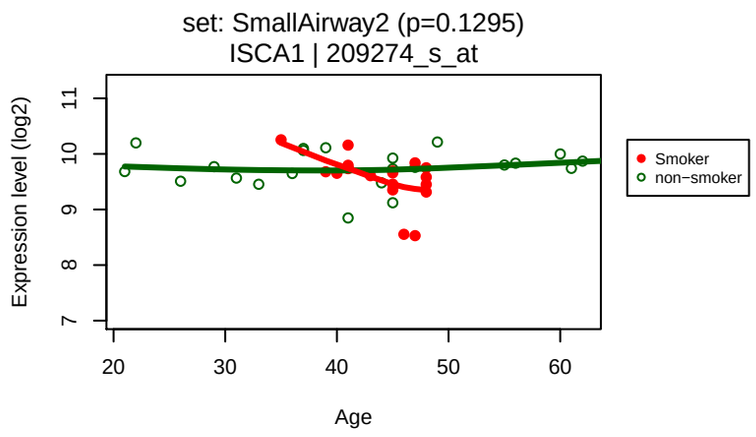
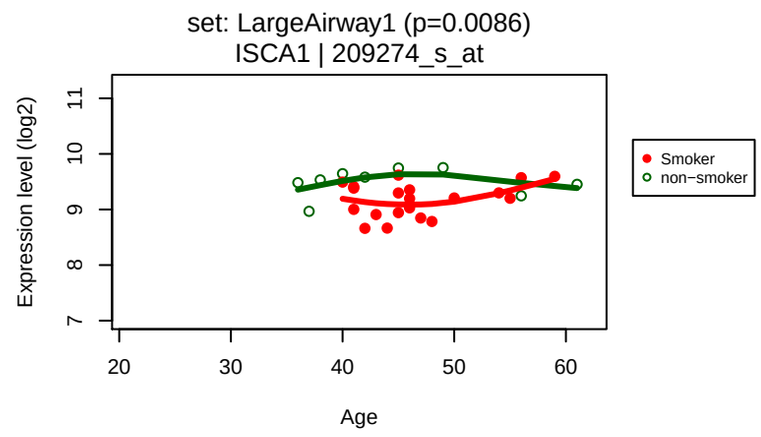

Gene expression of smokers and nonsmokers
gene: ISCA1 | 209274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0904
p=0.1295
p=0.0023
p=0.0086
p=0.2637
p=0.0039
p=0.7996
p=0.872
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

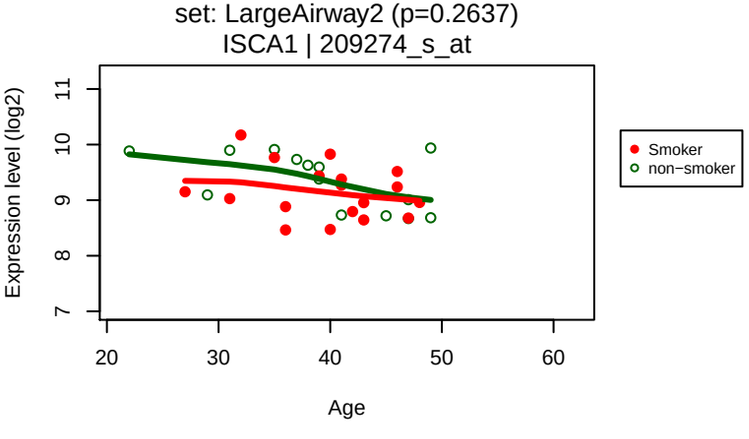

Gene expression ratio (smokers/nonsmokers)
gene: ISCA1 | 209274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0904
p=0.1295
p=0.0023
p=0.0086
p=0.2637
p=0.0039
p=0.7996
p=0.872
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

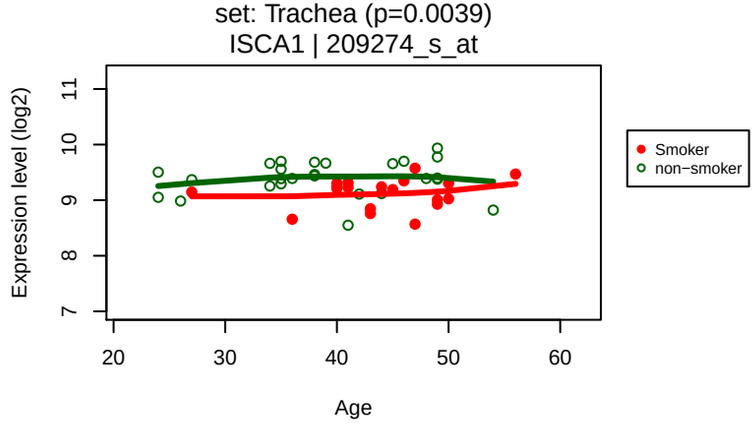

Gene expression ratio in Males
gene: ISCA1 | 209274_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7802
p=0.0518
p=0.0042
p=0.7829
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ISCA1 | 209274_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.141
p=0.3579
p=0.1399
p=0.907
SmallAirway2
LargeAirway2
Trachea
Macrophage2

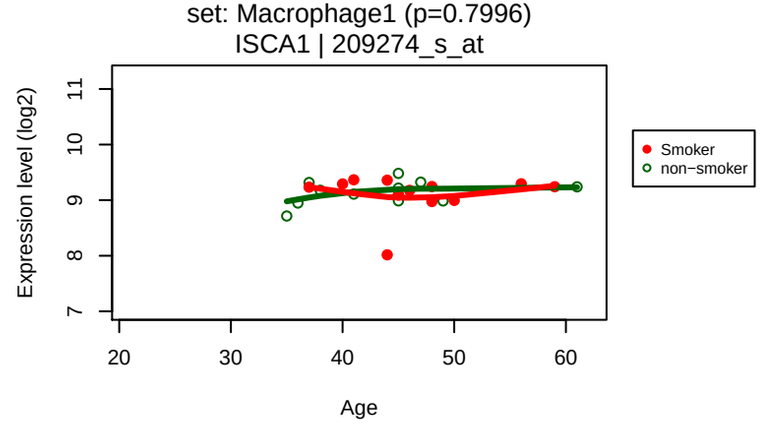

set: SmallAirway1 (p=0.0904)
ISCA1 | 209274_s_at
Expression level (log2)
Age
Smoker
non−smoker

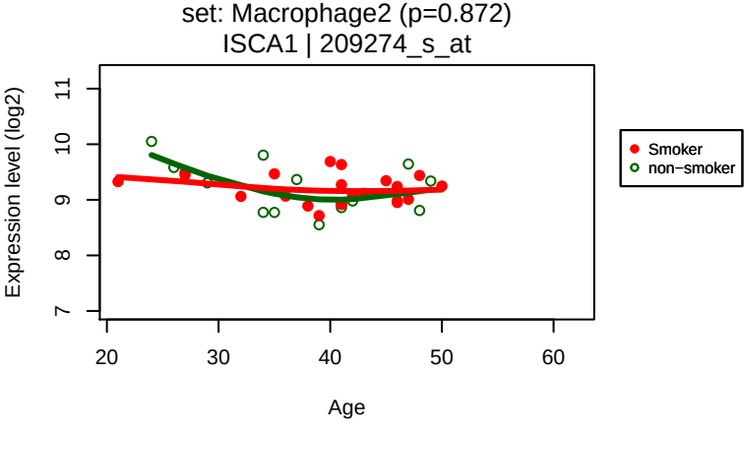

set: SmallAirway2 (p=0.1295)
ISCA1 | 209274_s_at
Expression level (log2)
Age
Smoker
non−smoker

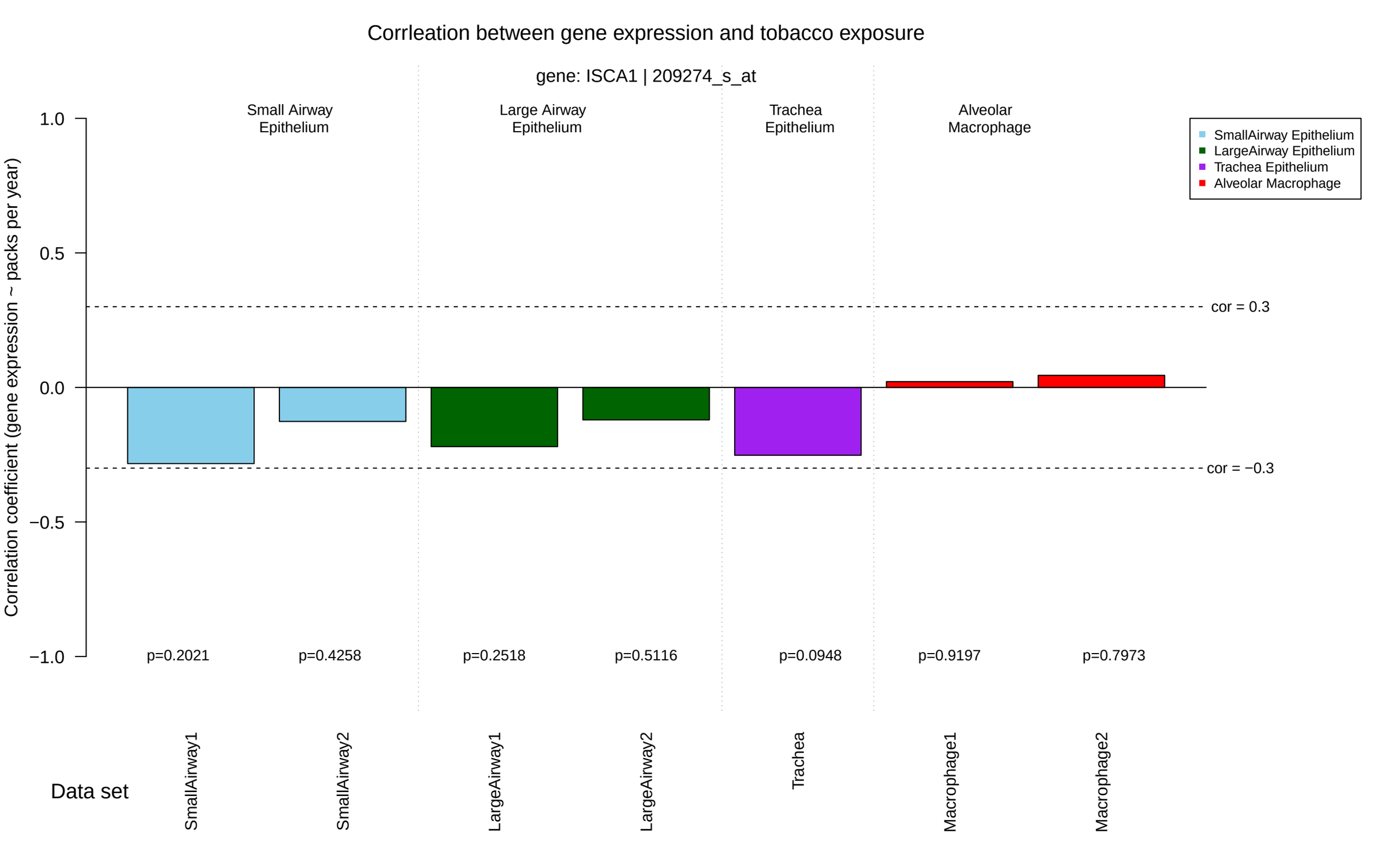

set: LargeAirway1 (p=0.0086)
ISCA1 | 209274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2637)
ISCA1 | 209274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0039)
ISCA1 | 209274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7996)
ISCA1 | 209274_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.872)
ISCA1 | 209274_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ISCA1 | 209274_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2021
p=0.4258
p=0.2518
p=0.5116
p=0.0948
p=0.9197
p=0.7973
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2